ScopeFun

User manual

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Introduction

ScopeFun in an open source, all-in-one, instrumentation platform. It combines several instruments: Oscilloscope, Arbitrary Waveform Generator, Spectrum Analyzer, Logic Analyzer, Digital Pattern Generator and more.

ScopeFun is designed for high performance, low noise operation and includes advanced technologies like FPGA and DDR3 SDRAM. Together with the free and open source software it provides a powerful and flexible test and measurement platform.

License

ScopeFun software, firmware and hardware design sources are available for free, all under open source licenses. Anyone can study the design, learn how it works and develop new applications based on the existing design.

ScopeFun project is licensed according to the following open source licenses:

- Software & Firmware is licensed under GNU General Public License version 3
- Hardware is licensed under CERN OHL version 1.2

Safety

Please read the safety information below before using ScopeFun for the first time.

Maximum input range

The full scale measurement ranges are the maximum voltages that can be accurately measured by each instrument. The overvoltage protection ranges are the maximum voltages that can be applied to the inputs without damaging the instrument.

ScopeFun is designed to measure voltages on **inputs CH1 and CH2** in the range from -10 V to +10 V. The CH1, CH2 inputs are protected for overvoltage to ±50 V.

Analog **outputs AWG1 and AWG2** can generate voltages up to ±2V and are protected against short circuit and overvoltage to ±25 V.

Digital general purpose channels (GPIO) are designed to accept input voltages from 0 V to +5 V and are protected from -5 V to +15 V. The logic 'High' voltage of digital GPIO channels can be adjusted in the range from 1.25 V to 3.3 V.

The following table summarizes voltage limits:

	Full scale range	Protection range
CH1, CH2 input	-10 V to +10 V	± 50 V
AGW1, AGW2 output	-2 V to +2 V	± 25 V
Digital GPIO **D11, D10, …,D0**	0 V to +3.3 V (+5 V)	-5 V to + 15 V

WARNING: ScopeFun is not designed to measure voltages outside the full scale range. Safety protection built in to the equipment may cease to function if input voltage limits are exceeded. To prevent electric shock, do not attempt to measure voltages outside the full scale range.

If the unit is used in a manner not specified by the manufacturer the degree of protection may be impaired.

Grounding

ScopeFun does not have a protective safety ground terminal. The ground connection to the computer ground through the USB cable is for measurement purposes only.

WARNING: Never connect the ScopeFun ground to any potential other than ground. To prevent personal injury or death, use a voltmeter to check that there is no significant AC or DC voltage between the ScopeFun ground and the point to which you intend to connect it.

RFI/EMI interference

Low voltage measurements can be susceptible to RFI (radio frequency interference) and EMI (electromagnetic interference) when performed in electrically noisy environments. **If the RFI/EMI interference is strong enough, it can be superimposed on the signal being measured.**

RFI/EMI interference can be minimized with different methods when taking sensitive measurements. A simple but effective method is to keep all instruments, cables, and DUTs as far from the interference source as possible. Another method is to enable signal filtering techniques on measured signals. Interference effects can also be reduced by shielding the test leads and the DUT.

Specifications

Oscilloscope

No. of channels	2 channels (CH 1, CH 2)
Bandwidth (-3 dB)	100 MHz
Sampling rate, max.	250 MSps dual channel 500 MSps single channel 2.0 GSps with Equivalent-Time Sampling (ETS)
Sampling rate, min.	50 Sps
Resolution	10 bit
Memory depth	128,000,000 samples per channel
Max. input voltage	20 V
Overvoltage protection	± 50 V
Input voltage ranges	± 10 V, ± 5 V, ± 2.5 V, ± 1 V, ± 500 mV, ± 250 mV, ± 100 mV, ± 50 mV
Input offset	adjustable
Input coupling	DC, AC, GND
Input impedance	1 MΩ || 18 pF

Arbitrary Waveform Generator

No. of channels	2 channels (AWG 1, AWG 2)
Sampling rate, max.	200 MSps
Resolution	12 bit
Output impedance	50 Ω
Output amplifier bandwidth	30 MHz
Waveform shapes	Sin, Cos, Triangle, Saw, Ramp up/down, Delta, DC, Noise, Custom
Custom waveform memory	32,768 Samples per Channel
Max out. voltage	± 2.0 V
Offset, level and phase	Adjustable
Overvoltage protection	Short-circuit, ± 25 V

Digital GPIO (Logic Analyzer, Digital Pattern Generator)

No. of channels	12 channels
Digital GPIO configuration options	12-input (Logic analyzer) OR 12-output (Digital pattern generator) OR 6-input, 6-output (Logic analyzer & Digital pattern generator)
Sampling rate max.	250 MSps
Interface voltage range	Adjustable from 1.25 V to 3.3 V in 256 steps
LA memory depth	128,000,000 samples per channel
Output impedance	526 Ω
Overvoltage protection	-5 V to +15 V
Digital pattern generator memory	32,768 samples per channel
Digital pattern generator internal clock divider	Adjustable 32-bit (250 Mhz - 0,058 Hz)

Trigger

Source	CH 1, CH 2, AWG 1, AWG 2, Digital GPIO
Mode	Auto, Normal, Single (with Re-Arm)
Pre-trigger	Adjustable 0 - 99%
Trigger level	0 - 100 %
Trigger hysteresis	Adjustable
Trigger hold-off	Adjustable 0 - 17 s (4 ns step size)
Digital trigger	4 stages (with delay counter for each stage)
Digital trigger	Selective channel masking (logic levels: '0', '1', 'Rising', 'Falling')

Environment

The unit is only for indoor use. Use the unit on a stable work surface that allows for good ventilation.

Environment	Residential
Operating temperature	0 °C to +38 °C
Altitude	2000 m
Maximum Relative humidity	80% up to 31 °C. Between 31 °C and 38 °C humidity linearly decreasing to 50 %

Cleaning

Before cleaning, disconnect the unit from computer. Wipe with a cloth soaked with alcohol.

Repairs

ScopeFun contains no user-serviceable parts. Repair of ScopeFun requires specialized test equipment and must be performed by the manufacturer.

Manufacturer details

ScopeFun
Address: Dejan Priversek, s.p.
Vicanci 97
SI-2274 Velika Nedelja
Slovenia
E-mail: info@scopefun.com
Website: www.scopefun.com

Design overview

Hardware

ScopeFun is an USB bus powered device, therefore it relies on the USB host to provide power via the USB cable. The unit is supplied from USB connector of PC. Personal computer shall be certified to IEC 60950-1 or IEC 62368-1.

For reliable operation, ScopeFun must be connected to the computer port supporting the **USB version 3.0**. The connecting USB cable must be of a high quality and extensions cables must not be used. If ScopeFun is connected to the USB port version 2.0, the USB port must be capable of providing at least 900 mA of current.

Software

ScopeFun software provides access to all ScopeFun instruments and hardware settings. When a setting is changed in the software, this is immediately reflected in the hardware.

ScopeFun software is available for all major operating systems (Windows, Linux, Mac OS) and includes:

- ***ScopeFun*** – main application which controls all the ScopeFun instruments
- ***Python API*** – Python scripting API interface which provides additional functionality through custom scripts

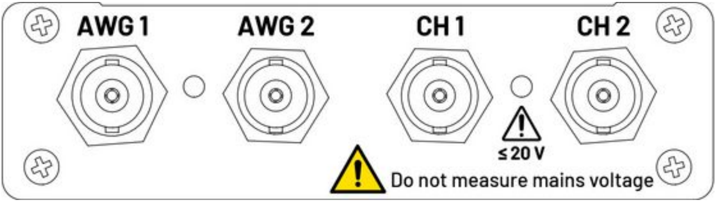
The latest version of ScopeFun software can be downloaded from the following web page: https://www.scopefun.com/download

Before first use

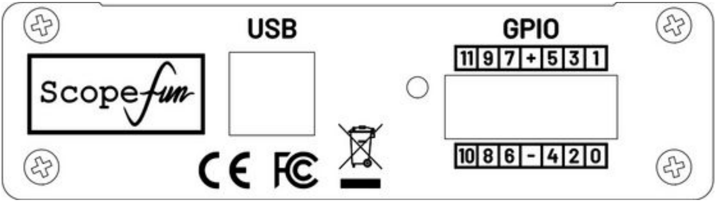
Important: Before using ScopeFun for the first time, please read the safety instructions on page 2.

Connectors

The following images show connector locations on the ScopeFun hardware.



Front Panel



Back panel

Driver

ScopeFun does not require manual driver installation. Drivers are automatically installed by the operating system when ScopeFun is connected for the first time.

Software

Please download and install the latest version of the ScopeFun software from the following web page: https://www.scopefun.com/download

Using the Software

Introduction

ScopeFun software is released under open source license GNU GPL.

Please note that new versions of the ScopeFun software are subsequently released, so instructions that follow may not be the latest. To check for the latest version of the this manual, please visit the following link: https://www.scopefun.com/download

Connecting to ScopeFun

Hardware is automatically detected after the software is launched.

The software will first search for ScopeFun board and if detected, the FPGA firmware image will be automatically uploaded. After successfull connection, the hardware calibration data will be read from the on-board EEPROM.

The following two check boxes indicate hardware status.

☒ USB Open	☒ FPGA
USB connection status	*FPGA firmware status*

To start the capture process, click on the 'Capture' button.

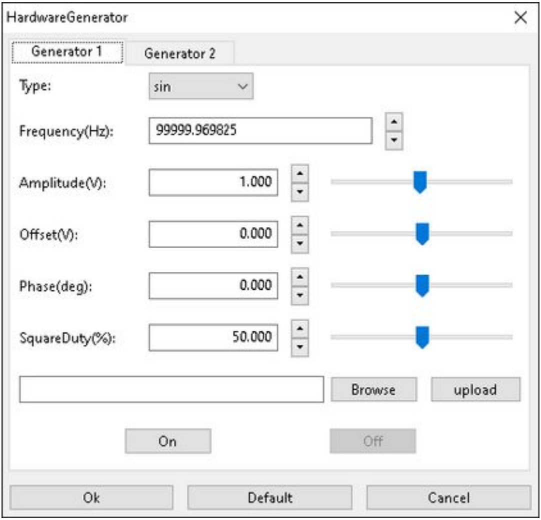
After performing the steps above, the software can be used to control all the ScopeFun instruments: Oscilloscope, Arbitrary Waveform Generator, Spectrum Analyzer, Logic Analyzer and Digital Pattern Generator.

Oscilloscope

All the oscilloscope settings can be controlled from the main software window. The ScopeFun oscilloscope instrument can be used like a traditional oscilloscope. User can adjust horizontal scale (voltage per division), vertical scale (sampling period) and offset (y-position). Trigger source, trigger level and trigger hysteresis can also be adjusted.

Arbitrary Waveform Generator (AWG)

Controls for the AWG instrument can be accessed from the menu, with the following selection: *Hardware -> Hardware Generator*.

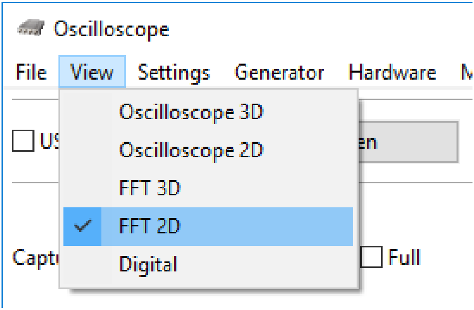


AWG settings

AWG settings include Frequency, Amplitude, Offset and Phase. The duty cycle of the rectangular waveform shape can also be adjusted.

Spectrum Analyzer

Spectrum analyzer (FFT view) can be enabled through the menu option: *View -> FFT 2D.*



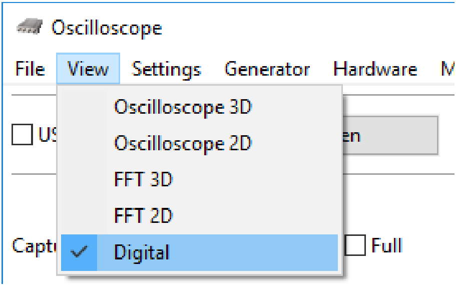
View FFT menu option

The option ‘Display FFT’ can be enabled for each oscilloscope input channel separately:

Enable FFT option

Logic Analyzer

Logic analyzer view can be enabled through the menu option: *View -> Digital.*



View Digital menu option

Individual logic analyzer channels can be enabled through the Digital channel tab on the main window.

Digital Pattern Generator

The digital pattern generator settings can be adjusted on the following two tabs: *Digital Setup* and *Digital Pattern.*

Digital Setup:

- Load custom digital pattern from file
- Adjust output HIGH voltage level
- Adjust output frequency of the digital pattern generator (Frequency divider)

Digital Pattern:

- Configure direction of the individual channel groups (Input / Output)
- Control individual digital channels:
  - **‘1’** - HIGH
  - **‘0’** - LOW
  - **‘X’** - CUSTOM BITS FROM FILE

Calibration

Although ScopeFun hardware is already calibrated at the manufacturer's site, it is recommended to periodically perform the self-calibration procedure to improve instrument accuracy.

Calibration procedure

Note: Before starting the calibration, make sure that ScopeFun has been acquiring samples for at least 15 minutes – this will allow the unit to warm up. It is important to calibrate the instrument at a temperature close to that at which it will be operated. If the operating temperature changes by more than 10° C, the calibration procedure should be repeated.

Calibration can be started by selecting "Auto Calibrate" from the software menu (the software must be launched with administrator privileges). To complete the calibration, please follow the on-screen instructions as indicated by the software (the process involves connecting CH1, CH2 inputs to AWG1, AWG2 outputs with BNC cable). The calibration procedure will automatically adjust the offset and gain of the CH1, CH2 inputs and offset of the AWG1, AWG2 outputs. After the calibration is finished, the calibration data will be saved to file: "./data/startup/hardware2.json". After the calibration is complete, the calibration data must also be saved to the ScopeFun hardware. This can be done manually by selecting "Write Calibration" from the software menu – this will copy the calibration data from the file "hardware2.json" to the on-board EEPROM.

Serial number

Each ScopeFun unit is given a unique serial number that is digitally signed and stored in EEPROM. To read the serial number and the digital signature from the connected ScopeFun board, select "Read Certificate" from the software menu. The serial number and the digital signature will then be copied to clipboard.

The digital signature is based on Ethereum Signatures and can be verified using tools such as:

- https://www.myetherwallet.com/tools?tool=verify

ScopeFun Ethereum address which was used to sign the message is:

- 0xa7a2b8F3de434AFAfe332480A95dD7C42D824500

Warranty and returns

ScopeFun is supplied with a two (2) year return-to-manufacturer limited warranty.

This warranty covers manufacturing defects and starts on the date of purchase. The warranty will be applied, if the product is properly handled and the mandatory preventive actions are followed as described in this manual.

This warranty does not cover any defects from unauthorized modification or alteration of the original condition. The warranty is void if malfunction or damage is caused by misuse according to warnings in the product documentation or using or storing in an environment outside of the specification’s limits. The warranty and financial responsibility of the manufacturer only applies to the product itself and does not extend to items that might be used with the purchased product, except where mandated by law.

To obtain a repair or replacement under the terms of this warranty, please contact the manufacturer first to obtain a returns number, and the manufacturer will direct you accordingly for repairs or replacement. You will be required to submit a copy of the original receipt.

Conformance

ScopeFun has been designed, tested and found to comply with international Electromagnetic Compatibility (EMC) and safety standards:

- IEC 61326-1:2012 (Second Edition) - Electrical equipment for measurement, control and laboratory use – EMC requirements
- Title 47 CFR Part 15, Subpart B (Clause 15.107 and 15.109) in conjunction with ANSI C63.4:2014
- Fast evaluation, clauses 5.1, 5.2 and 5.4 of IEC 61010-1:2010 + A 1: 2016, EN 61010-1:2010 + A 1 :2019, clauses 5.1 .5, 5.4.1 and 5.4.3 of IEC 61010-2- 030:201 O; EN 61010-2-030:2010

FCC notice

This equipment has been tested and found to comply with the limits for a **Class B** digital device, pursuant to **part 15 of the FCC Rules**. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

Reorient or relocate the receiving antenna.

Increase the separation between the equipment and receiver.

Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.

Consult the dealer or an experienced radio/TV technician for help.

CE notice

The product meets the intent of the following European Union directives: **2014/30/EU (EMC), 2012/19/EU (WEEE) and 2015/863/EU (RoHS)**

Waste Electrical and Electronic Equipment (WEEE) Directive

figure 1: WEEE symbol – crossed out wheeled bin

For private households: Information on Disposal for Users of WEEE

This symbol (figure 1) on the product(s) and / or accompanying documents means that used electrical and electronic equipment (WEEE) should not be mixed with general household waste. For proper treatment, recovery and recycling, please take this product(s) to designated collection points where it will be accepted free of charge. Alternatively, in some countries, you may be able to return your products to your local retailer upon purchase of an equivalent new product.

Disposing of this product correctly will help save valuable resources and prevent any potential negative effects on human health and the environment, which could otherwise arise from inappropriate waste handling.

Please contact your local authority for further details of your nearest designated collection point. Penalties may be applicable for incorrect disposal of this waste, in accordance with you national legislation.

For professional users in the European Union

If you wish to discard electrical and electronic equipment (EEE), please contact your dealer or supplier for further information.

For disposal in countries outside of the European Union

This symbol is only valid in the European Union (EU). If you wish to discard this product please contact your local authorities or dealer and ask for the correct method of disposal.